

Scientific Computation

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Lecture 14 - Symmetry and Positive Definiteness

Lecture Outline

- Symmetric Matrices

- Positive Definiteness
- The Cholesky Decomposition

Symmetric Matrices

Consider a **symmetric matrix** $A \in \mathbb{S}^n$, i.e., $A^\top = A$, and suppose that it has an LU decomposition $A = LU$.

Instead of n^2 possible unique scalar entries, we have at most

$$\frac{n(n+1)}{2} \approx \frac{n^2}{2}.$$

Can we exploit this fact to efficiently solve $Ax = b$ with a factorization?

2-by-2 Case

Example 1 (Gath and van Leeuwen): Consider a general 2-by-2 symmetric

Example 1 (Golub and van Loan): Consider a general 2-by-2 symmetric matrix:

$$A = \begin{bmatrix} a & c \\ c & d \end{bmatrix}.$$

The LU factorization is:

$$L = \begin{bmatrix} 1 & 0 \\ \frac{c}{a} & 1 \end{bmatrix} = U = L^{-1}A = \begin{bmatrix} a & c \\ 0 & d - \frac{c^2}{a} \end{bmatrix}$$

$$\begin{aligned} LU &= \begin{bmatrix} 1 & 0 \\ \frac{c}{a} & 1 \end{bmatrix} \begin{bmatrix} a & c \\ 0 & d - \frac{c^2}{a} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ \frac{c}{a} & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d - \frac{c^2}{a} \end{bmatrix} \begin{bmatrix} 1 & \frac{c}{a} \\ 0 & 1 \end{bmatrix} \\ &= LDL^T, \text{ D is diagonal} \end{aligned}$$

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LDL^T Factorization

Theorem: If $A \in \mathbb{S}^n$ and it has a unique LU factorization, then there exists a lower unit triangular matrix L and a diagonal matrix D such that

$$A = LDL^T. \quad L^{-1} = (L^T)^T$$

$$A = LU, \text{ Consider } D \triangleq L^{-1}AL^{-1} = UL^{-T}$$

$$\text{where } L^{-T} \triangleq (L^T)^{-1} = (L^{-1})^T$$

First: product of upper / lower tri. matrices is upper / lower triangular

$$(LL^T)_{ij} = \sum_{k=1}^n l_{ik}l'_{kj}, \text{ now suppose } j > i. \text{ Then, } l_{ik} = 0 \text{ when } k > i,$$

$$\text{but } l'_{kj} = 0 \text{ when } k \leq i \leq j.$$

so, LL' is also lower triangular

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LDL^T Factorization cont.

The product of lower triangular matrices remaining lower triangular also gives us insight about their inverses:

$$L = \begin{bmatrix} l_{11} & & 0 \\ \vdots & \ddots & \\ l_{n1} & \dots & l_{nn} \end{bmatrix} \rightarrow \text{When we do } GL, \text{ it remains upper triangular and becomes diagonal.}$$

$$M_{n-1} M_{n-2} \dots M_1 L = D \rightarrow M = DL^{-1}, \text{ and } L^{-1} = D^{-1}M \rightarrow \text{lower triangular} \\ \hookrightarrow ML$$

$\therefore L^{-1}$ is the product of lower triangular matrices.

L^{-1} is also lower triangular

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LDL^T Factorization Cont.

Now that we know L^{-1} is also lower diagonal, we can see that:

$D = L^{-1}AL^{-1} = UL^{-1}$ is the product of upper triangular matrices and therefore upper triangular.

$$\text{But } D \text{ also symmetric: } (L^{-1}AL^{-1})^T = (L^{-1})^T A^T (L^{-1})^T \\ = L^{-1}AL^{-1}$$

Therefore D is upper triangular and symmetric:

$$D = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix}, \quad D_{uu} = 0 \rightarrow D \text{ is diagonal}$$

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Algorithm for LDL^T

We can use the symmetric structure of the factorization to derive the following equations for the elements $d \triangleq \text{diag}(D)$:

$$A = \underbrace{L}_{L.T.} \underbrace{DL^T}_{L.T.} = L \begin{bmatrix} d_1 l_{11} & d_1 l_{12} & \cdots & d_1 l_{1n} \\ & d_2 l_{22} & & \vdots \\ & & \ddots & \\ & & & d_n l_{nn} \end{bmatrix}$$

$$\begin{aligned} \rightarrow a_{ii} &= (LDL^T)_{ii} = \sum_{k=1}^n l_{ik} d_k l_{ik} \\ &= \sum_{k=1}^{i-1} (l_{ik})^2 d_k + l_{ii} d_i \\ &= \sum_{k=1}^{i-1} (l_{ik})^2 d_k + d_i \end{aligned}$$

$$d_i = a_{ii} - \underbrace{\sum_{k=1}^{i-1} (l_{ik})^2 d_k}_{\text{Relies on previous } d_k \text{ and } l_{ik}}$$

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Algorithm for LDL^T Cont.

With our equation for the diagonal, we can determine the elements of L as follows:

$$A = LDL^T \rightarrow L.T. \quad l_{jk} = 0 \text{ for } k > j$$

$$\begin{aligned} \rightarrow a_{ij} &= (LDL^T)_{ij} = \sum_{k=1}^n l_{ik} d_k l_{jk} \\ &= \sum_{k=1}^{j-1} l_{ik} d_k l_{jk} + l_{ij} d_j l_{jj} \end{aligned}$$

$$\rightarrow l_{ij} = \frac{1}{d_j} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} d_k l_{jk} \right)$$

Efficiency of LDL^T

In addition to the memory savings, computing this symmetric factorization is about twice as computationally efficient:

$$A = \begin{bmatrix} a_{11} & \cdots & L^T \\ L & & a_{nn} \end{bmatrix}, \quad \text{so only store half of elements.}$$

Lecture Outline

- Symmetric Matrices
- **Positive Definiteness**
- The Cholesky Decomposition

Positive (Semi-)Definite Matrices

A matrix $A \in \mathbb{R}^{n \times n}$ is **positive definite** if

$$x^T A x > 0$$

for all nonzero $x \in \mathbb{R}^n$.

we write $A \succ 0$ or say $A \in \mathcal{SS}_{++}^n$ if A is also symmetric

Similarly, A is positive semi-definite if $x^T A x \geq 0 \quad \forall x$,
 notation $A \geq 0$, $A \in \mathcal{SS}_+^n$ for symmetric.

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Properties of P(S)D Matrices

For $A \in \mathcal{S}_{++}^n$, we have that:

A is invertible. We can see this by noting that $Ax=0$ iff $x=0$
 ($Ax=0 \rightarrow x^T Ax=0 \rightarrow x=0$). Therefore, kernel of A has dimension 0
 $\Leftrightarrow A$ is invertible.

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Examples of P(S)D Matrices

Symmetric PD and PSD matrices are extremely common:

stats : $\text{Cov}(x) = \mathbb{E}[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T]$

which has sample covariance matrix $\Sigma = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T$

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Eigenvalues and Symmetry

The characteristic equation of $A \in \mathbb{R}^{n \times n}$ is a polynomial of degree n , which means it may have complex solutions. However, for real **symmetric** matrices we have

Let $v \in \mathbb{C}^n$ with eigenvalue $\lambda \in \mathbb{C}$. For any $x \in \mathbb{C}^n$, we have

$$\begin{aligned} \bar{x}^T x &= \sum_{j=1}^n (a_j - b_j i)(a_j + b_j i) \\ &\stackrel{\text{complex conjugate}}{=} \sum_{j=1}^n a_j^2 + b_j^2 \geq 0 \end{aligned}$$

$$\text{let } x = Av = \lambda v$$

$$\begin{aligned} \rightarrow \bar{x}^T x &= \|x\|^2 = (\bar{Av})^T Av \\ &= \bar{v} \bar{A}^T A v = \bar{v} A^2 v \\ &= \lambda^2 \|v\|^2 \end{aligned}$$

$$\rightarrow \lambda^2 = \frac{\|x\|^2}{\|v\|^2} \geq 0 \rightarrow \text{all eigenvalues are real}$$

$$\|v\|^2 = v^T v$$

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Eigenvalues and Positive (Semi-)Definiteness

There is another useful characterization of PD and PSD matrices:

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Lecture Outline

- Symmetric Matrices
- Positive Definiteness

▼ POSITIVE DEFINITENESS

- The Cholesky Decomposition

The Cholesky Factorization

The symmetric LDL^T factorization has a special case when $A \succ 0$:

关于 positive definite和semi definite matrix:

设 A 是一个 **实对称矩阵** (即 $A=A^T$) 。

考虑一个二次型 (quadratic form) :

$$q(x)=x^T A x$$

其中 $x \in \mathbb{R}^n$

矩阵 A 的「正定性」就是通过 $q(x)$ 的符号来定义的。

所有PD矩阵都是invertible的！ 因为定义 $x^T A x > 0$ ，不可能出现 $Ax=0$ 的情况。

✓ 1. Positive Definite (正定矩阵)

矩阵 A 是 **正定的**，当且仅当：

$$x^T A x > 0, \quad \forall x \neq 0$$

也就是说：

- 除了 $x = 0$ 以外，任意向量代入 $x^T A x$ 都得到正数；
- 几何意义上：二次型 $x^T A x$ 定义了一个“碗形向上”的曲面。

符号记法：

$A \succ 0$ 表示 A 是正定的。



● 2. Positive Semi-definite (半正定矩阵)

矩阵 A 是 **半正定的**，当且仅当：

$$x^T A x \geq 0, \quad \forall x$$

区别在于这时 **可以等于 0**，即某些方向 x 上的“能量”为零。

符号记法：

$A \succeq 0$ 表示 A 是半正定的。

● 三、从特征值角度理解

因为对称矩阵可以被正交对角化：

$$A = Q \Lambda Q^T$$

其中 Q 是正交矩阵（旋转变换）， $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ 是特征值对角矩阵。

那么：

$$x^T A x = (Q^T x)^T \Lambda (Q^T x) = \sum_{i=1}^n \lambda_i (Q^T x)_i^2$$

$$x^T Ax = (Q^T x)^T \Lambda (Q^T x) = \sum_i \lambda_i y_i^2$$

其中 $y = Q^T x$ 。

由此：

类型	定义条件	特征值条件	记法	
正定 (PD)	$x^T Ax > 0$ 对所有非零 x	所有特征值 > 0	$A \succ 0$	
半正定 (PSD)	$x^T Ax \geq 0$ 对所有 x	所有特征值 ≥ 0	$A \succeq 0$	
负定 (ND)	$x^T Ax < 0$ 对所有非零 x	所有特征值 < 0	$A \prec 0$	
半负定 (NSD)	$x^T Ax \leq 0$ 对所有 x	所有特征值 ≤ 0	$A \preceq 0$	
不定 (Indefinite)	$x^T Ax$ 有时正、有时负	有正有负的特征值	—	

那么它的几何性质是怎么样子的：

当你有一个对称矩阵 A ，我们可以定义一个函数

$$f(x,y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix} = x^T Ax$$

这个函数的图像在三维空间中是一个**曲面**（类似山谷、平面或马鞍）。

● 1. 正定矩阵（Positive definite）→ “碗形向上”

例子：

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow f(x,y) = 2x^2 + 3y^2$$

图像是一个**碗形向上**的曲面。

- 不论你往哪个方向走， $f(x,y)$ 都会增加（因为 x^2 和 y^2 永远是正的）。
- 唯一的最低点在原点 (0,0)。

➡ 所以说它“碗形向上”，说明它在所有方向都是弯上去的。
这和二次函数 $y = x^2$ 是同一个形状，只是变成了二维的碗。

● 2. 半正定矩阵 (Positive semi-definite) → “平碗”

例子：

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow f(x, y) = x^2$$

图像像一个**平底碗**：

- 沿着 x -轴方向是弯上去的；
- 但沿着 y -轴方向是**完全平的**，因为 y 不会影响 $f(x, y)$ 。

➡ 所以称它是“平碗”，有些方向（比如 y 方向）是平的。

● 3. 不定矩阵 (Indefinite) → “马鞍形”

例子：

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow f(x, y) = x^2 - y^2$$

这个图像是个**马鞍形 (saddle)**：

- 沿着 x 方向 (x 轴) 是弯上去的；
- 但沿着 y 方向是弯下去的。

➡ 所以函数在某些方向增加，在另一些方向减少——不是“碗”，而是“马鞍”。